

The Weighted-Ellipsoid Determinant Trace Is Exactly 2/3

Partial-result packet for arXiv:2012.11115

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Status: candidate substantial partial result, likely valid; human review requested. For every $\lambda \geq 4$, the coordinate-multiplication pair on the weighted Bergman space of the source ellipsoid $\mathbb{B}_{2,1} = \{(z_1, z_2) : |z_1|^2 + |z_2| < 1\}$ has determinant-commutator trace exactly 2/3. This converts the source's numerical evidence into a proof and verifies its spectral-volume conjecture with equality for its only non-ball model. The conjecture for arbitrary tuples remains open.

1 Source conjecture and numerical test case

The source is G. Misra, P. Pramanick, and K. B. Sinha, *A trace inequality for commuting tuple of operators*, arXiv:2012.11115 [1]. For a commuting d -tuple $\mathbf{T} = (T_1, \dots, T_d)$ it sets $C_{ij} = [T_j^*, T_i]$ and defines a symmetrized determinant $\text{dEt}(C)$. In two variables,

$$\text{dEt}(C) = C_{11}C_{22} + C_{22}C_{11} - C_{12}C_{21} - C_{21}C_{12}. \quad (1)$$

For tuples in the source class $BS_{m,\vartheta}(\Omega)$, this operator is positive and trace class. Conjecture 5.6 predicts

$$\text{tr dEt}([[\mathbf{T}^*, \mathbf{T}]]) \leq \frac{md!}{\pi^d} \nu(\overline{\Omega}). \quad (2)$$

The principal non-ball example is the pair $M = (M_{z_1}, M_{z_2})$ on weighted Bergman spaces $\mathbb{A}^{(\lambda)}(\mathbb{B}_{2,1})$. The source proves $M \in BS_{1,2}(\mathbb{B}_{2,1})$ for $\lambda \geq 4$, but states that it cannot compute the trace explicitly and reports numerical value 2/3.

Remark 5.5. Although, unlike the case of the Euclidean ball $\mathbb{B}_2$, we are not able to explicitly compute the trace of the operator $\text{dEt}([[\mathbf{T}^*, \mathbf{T}]])$ for the weighted Bergman spaces $\mathbb{A}^{(\lambda)}(\mathbb{B}_{2,1})$, extensive numerical computations show that it is approximately equal to $\frac{2}{3}$, which is the $\frac{2}{\pi^2}$ times the volume of the ellipsoid $\mathbb{B}_{2,1}$.

Figure 1: Source Remark 5.5 reports only numerical evidence for 2/3 (PDF page 28).

operator $\text{dEt}(\llbracket \mathbf{T}^*, \mathbf{T} \rrbracket)$. In the example of weighted Bergman spaces over the Euclidean ball $\mathbb{B}_d$, from Corollary 5.3, it follows that

$$\text{trace}(\text{dEt}(\llbracket \mathbf{T}^*, \mathbf{T} \rrbracket)) = \frac{md!}{\pi^d} \nu(\mathbb{B}_d).$$

Also, for the ellipsoid $\mathbb{B}_{2,1}$, we have numerical evidence for such an equality. Taking all of this into account, we make the following conjecture.

Conjecture 5.6. Suppose that the commuting d -tuple of operators $\mathbf{T} = (T_1, \dots, T_d)$ is in the class $BS_{m,\theta}(\Omega)$. Then

$$\text{trace}(\text{dEt}(\llbracket \mathbf{T}^*, \mathbf{T} \rrbracket)) \leq \frac{md!}{\pi^d} \nu(\overline{\Omega}),$$

where ν is the Lebesgue measure.

We have made some progress towards finding a solution to this conjecture.

Figure 2: The sharp spectral-volume bound in source Conjecture 5.6 (PDF page 29).

2 The model and result

For $\lambda > 3$, the source defines the inner product by the measure

$$(1 - |z_1|^2 - |z_2|^2)^{\lambda-4} d\nu(z) \quad \text{on} \quad \mathbb{B}_{2,1} = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 < 1\}.$$

The normalized monomials $\phi_{m,n}$, $m, n \geq 0$, form an orthonormal basis. Put $s = m + 2n + \lambda$. The source's norm formula gives

$$M_{z_1} \phi_{m,n} = \sqrt{\frac{m+1}{s}} \phi_{m+1,n}, \quad M_{z_2} \phi_{m,n} = \sqrt{\frac{(2n+2)(2n+3)}{s(s+1)}} \phi_{m,n+1}. \quad (3)$$

Theorem 1 (Exact ellipsoid trace). *For every $\lambda \geq 4$, let $M = (M_{z_1}, M_{z_2})$ act on $\mathbb{A}^{(\lambda)}(\mathbb{B}_{2,1})$. Then*

$$\text{tr dEt}(\llbracket M^*, M \rrbracket) = \frac{2}{3} = \frac{2!}{\pi^2} \nu(\mathbb{B}_{2,1}).$$

Consequently, source Conjecture 5.6 holds with equality for this entire one-parameter family.

3 Diagonalization

Write $C_{ij} = [M_{z_j}^*, M_{z_i}]$ and $D = \text{dEt}(C)$. The two diagonal commutators in (1) have respective eigenvalues

$$c_{11} = \frac{2n + \lambda - 1}{s(s-1)}, \quad c_{22} = \frac{(2n+2)(2n+3)}{s(s+1)} - \frac{(2n)(2n+1)}{(s-2)(s-1)}. \quad (4)$$

The mixed commutators move one step along the diagonals of the lattice. The squared coefficients contributing to $C_{12}C_{21}$ and $C_{21}C_{12}$ are

$$p^2 = \frac{4(m+1)(2n)(2n+1)}{(s-1)s^2(s-2)^2}, \quad q^2 = \frac{4m(2n+2)(2n+3)}{s(s+1)^2(s-1)^2}. \quad (5)$$

Thus D is diagonal and

$$D\phi_{m,n} = \chi_{m,n}\phi_{m,n}, \quad \chi_{m,n} = 2c_{11}c_{22} - p^2 - q^2. \quad (6)$$

Expanding (6) agrees identically with the four-term formula for χ_α printed on source page 28. The source proves $D \geq 0$ for $\lambda \geq 4$ [1, Theorem preceding Remark 5.5], so all $\chi_{m,n}$ are nonnegative.

4 Exact telescoping

Group the lattice by the weighted degree $m + 2n$. For $r \geq 0$, set

$$B_r = \sum_{n=0}^r \chi_{2r-2n,n} + \sum_{n=0}^r \chi_{2r+1-2n,n}. \quad (7)$$

We now give a rational certificate for the sum. Put $x_r = \lambda + 2r$ and

$$\begin{aligned} a &= -\frac{(\lambda-4)(\lambda-2)(\lambda^2-8\lambda+18)}{24}, & a_0 &= -\frac{(\lambda-2)(2\lambda^2-11\lambda+18)}{6}, \\ b &= -\frac{(\lambda-3)(\lambda-1)(\lambda^2-6\lambda+11)}{6}, & c &= \frac{3\lambda^4-30\lambda^3+96\lambda^2-88\lambda-28}{24}, \\ c_0 &= -\frac{3\lambda^4-30\lambda^3+96\lambda^2-96\lambda-8}{12}, & e &= \frac{2(2\lambda-5)}{3}. \end{aligned} \quad (8)$$

Define

$$F_r = \frac{a}{(x_r-2)^2} + \frac{a+a_0}{x_r^2} + \frac{b}{(x_r-1)^2} + \frac{c}{x_r-2} + \frac{c+c_0}{x_r} + \frac{e}{x_r-1}. \quad (9)$$

Lemma 2 (Telescoping certificate). *For every $r \geq 0$ and $\lambda \geq 4$,*

$$B_r = F_r - F_{r+1}, \quad F_0 = \frac{2}{3}, \quad \lim_{r \rightarrow \infty} F_r = 0.$$

Proof. Substitute $m = 2r - 2n$ and $m = 2r + 1 - 2n$ into (4)–(6), sum the resulting rational functions for $0 \leq n \leq r$, and collect the partial fractions with denominators $x_r + j$, $-2 \leq j \leq 2$. The squared-denominator coefficients at $x_r - 2, x_r, x_r + 2$ sum to zero, as do the first-order coefficients; the odd terms occur in opposite pairs at $x_r - 1, x_r + 1$. The result is exactly the forward difference of (9). Direct substitution and cancellation give $F_0 = 2/3$. Each paired first-order term in (9) cancels at infinity, and the remaining expression is $O(r^{-1})$, so $F_r \rightarrow 0$.

All three identities are exact rational identities. The accompanying SymPy script expands (6), checks it against the source formula, performs the two finite sums in (7), and verifies $B_r - (F_r - F_{r+1}) = 0$, $F_0 - 2/3 = 0$, and $\lim F_r = 0$ symbolically. $\square$

Proof of Theorem 1. Because $D \geq 0$, Tonelli's theorem permits the grouping (7). The certificate therefore yields

$$\text{tr } D = \sum_{m,n \geq 0} \chi_{m,n} = \sum_{r=0}^{\infty} (F_r - F_{r+1}) = F_0 = \frac{2}{3}.$$

Finally, polar coordinates give

$$\nu(\mathbb{B}_{2,1}) = 4\pi^2 \int_0^1 r_1 \int_0^{1-r_1^2} r_2 dr_2 dr_1 = \frac{\pi^2}{3}.$$

Hence $(2!/\pi^2)\nu(\mathbb{B}_{2,1}) = 2/3$, as claimed. $\square$

5 Scope, verification, and novelty

The theorem resolves the exact computation highlighted in source Remark 5.5, but it does not prove (2) for every tuple in $BS_{m,\vartheta}(\Omega)$. Eight upgrade attempts examined affine covariance, sums and tensor products, cyclic jet extensions, general Helton–Howe formulas, proper monomial maps, and discrete Monge–Ampère interpretations. The main obstruction is that the BS hypotheses control only the top-degree generalized commutator; they do not give the pairwise Schatten estimates or a bounded-multiplicity principal function needed by known trace-volume formulas.

Verification notes. Review should focus on (i) the weight calculation (3), (ii) the mixed-shift coefficients (5), and (iii) the finite rational identity in the telescoping lemma. The symbolic script is a certificate, not a numerical experiment: all checks occur over the rational-function field $\mathbb{Q}(\lambda, r)$. The use of nonnegative eigenvalues and Tonelli follows from the source’s $\lambda \geq 4$ positivity theorem.

Novelty check. On 11 August 2026, the four run indexes were searched for arXiv:2012.11115, “determinant trace”, “ellipsoid”, and “2/3”. Exact-title and exact-formula web/arXiv searches found no proof or counterexample. Most significantly, Pramanick’s revised 2025 thesis restates the bound as Conjecture 3.19 and still describes the ellipsoid value as numerical evidence [2]. Later weighted-Bergman Helton–Howe formulas found in the search concern the unit ball, not this branched ellipsoid model [3]. Novelty confidence is moderate-high.

Human-review recommendation. Review as a substantial partial result. It rigorously settles the source’s only non-ball test family and confirms the sharp constant, while making no claim that the full $BS_{m,\vartheta}$ conjecture is solved.

References

- [1] G. Misra, P. Pramanick, and K. B. Sinha, *A trace inequality for commuting tuple of operators*, arXiv:2012.11115, Theorem 5.4, Remark 5.5, and Conjecture 5.6.
- [2] P. Pramanick, *Trace Estimate for the Determinant Operator and Generalized Commutator*, revised doctoral thesis, Indian Institute of Science, 2025, Conjecture 3.19.
- [3] X. Tang, Y. Wang, and D. Zheng, *Helton–Howe trace, Connes–Chern character and quantization*, arXiv:2204.04337.
- [4] J. W. Helton and R. E. Howe, *Traces of commutators of integral operators*, Acta Math. 135 (1975), 271–305.